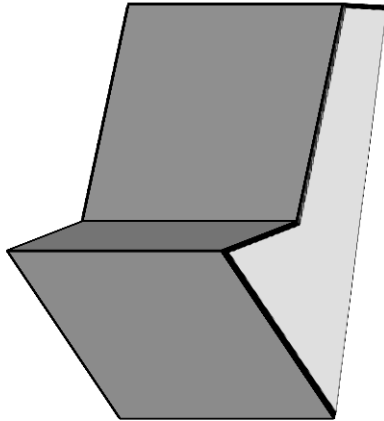


Graphical Processing Systems

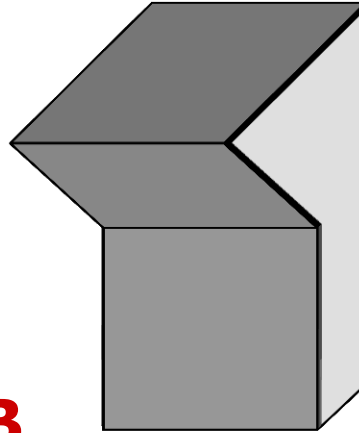
Examination Subjects

Figure 1. 3D Objects

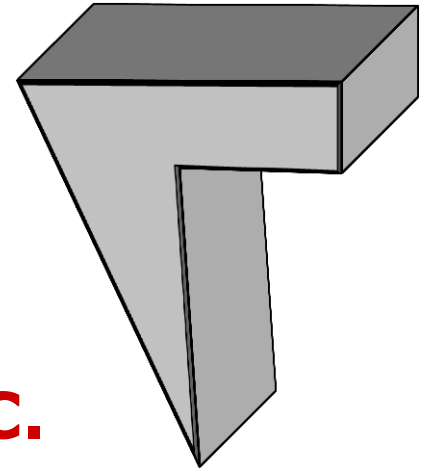
1A.



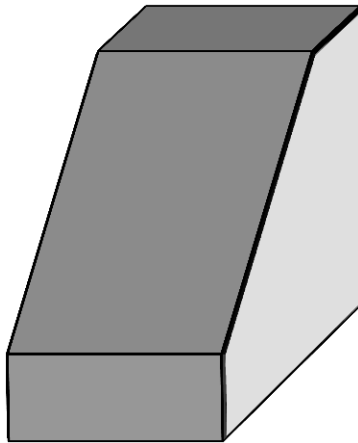
1B.



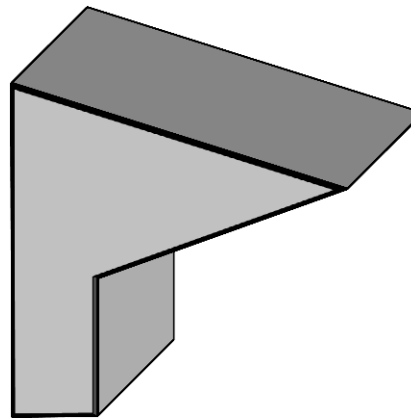
1C.



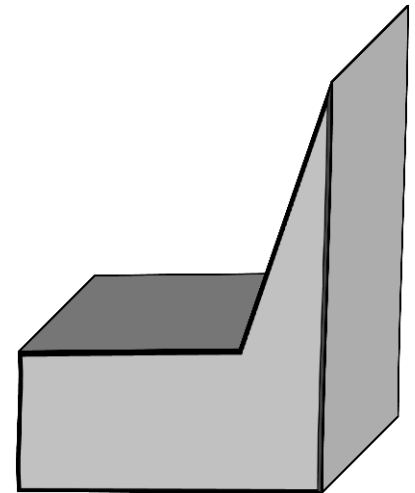
2A.



2B.



2C.



Subject 1

Exemplify and explain the visualization of the solid 3D object by BSP Tree, if the resulted image looks like in the figure 1.

Subject 2

Describe and explain a method to determine if the 3D object in the figure 1 is convex or concave.

Subject 3

Exemplify and explain for the complex 3D object in the figure 1, the following algorithm for the hidden line and surface removal. The resulted image by orthogonal projection is like in the figure.

1A. Scan-line

1B. Area subdivision: Warnock

1C. Depth sorting

2A. Depth sorting

2B. Scan-line

2C. Area subdivision: Warnock

Subject 4

Let us consider one light source and the 3D solid object of figure 1 defined in the graphics model only by its vertices (x, y, z) . Explain the computation of required parameters to do:

1A. Flat illumination by Phong reflection model

1B. Gouraud shading

1C. Phong shading

2A. Phong shading

2B. Flat illumination by Phong reflection model

2C. Gouraud shading

Subject 5

Explain the perspective transformation from view space into screen space highlighting the advantages and disadvantages of this transformation.

Subject 6

Let us consider the 3D solid object of figure 1 defined in the graphics model only by its vertices (x, y, z) .

Explain the computation of the two phase mapping of a given rectangular texture, and the main issues, if the intermediate surface is:

1A. Cube

1B. Cylinder

1C. Sphere

2A. Sphere

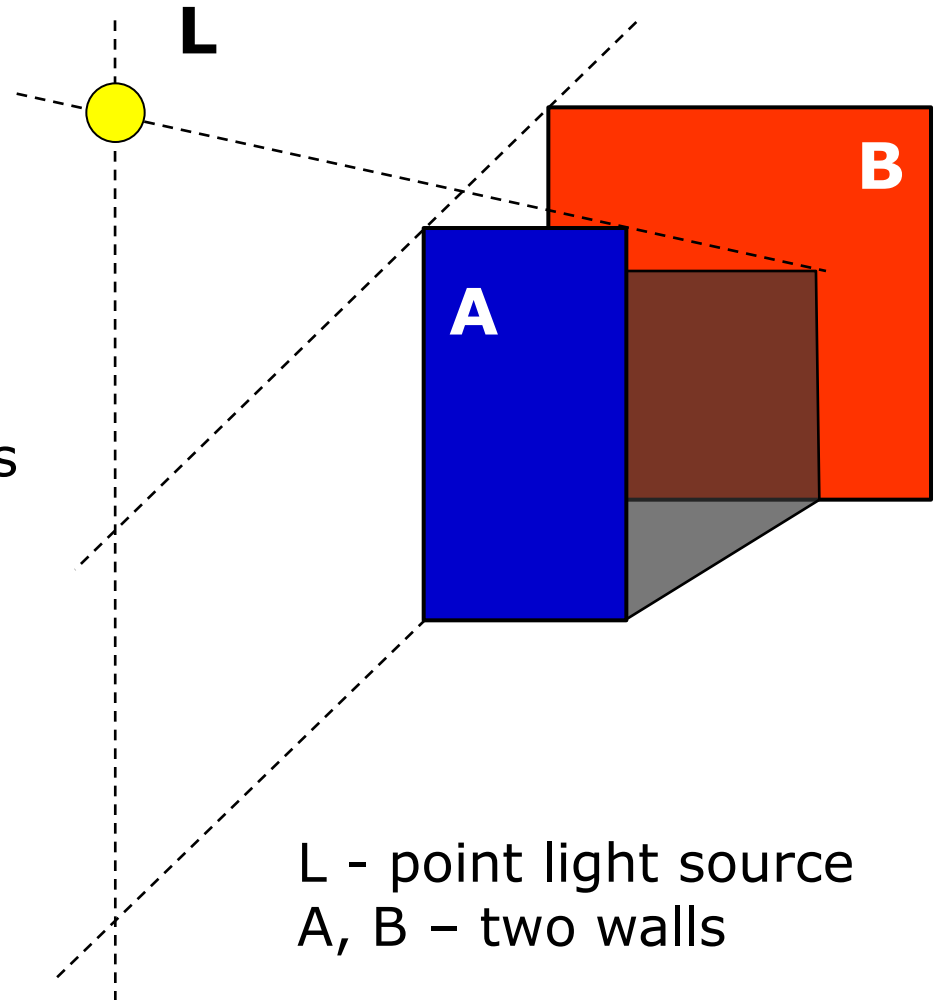
2B. Cube

2C. Cylinder

Subject 7

Explain the following shadow computation algorithm on the figure:

- 1A.** Shadows projected onto the ground plane
- 1B.** Scan line algorithm with integrating shadow polygons
- 1C.** Shadow volumes
- 2A.** Derivation of shadow polygons from light source transformations
- 2B.** Shadow Z-Buffer
- 2C.** Shadow volumes



Subject 8

Explain and exemplify each phase of the GPU graphics pipeline.

Subject 9

Let us consider the polygonal faces of the 3D object in figure 1, as distinct objects. Explain the parallel rendering by a graphics cluster working with

(Number of processors/Rendering algorithm):

1A. 3/Sort first

1B. 3/Sort last

1C. 4/Sort first

2A. 4/Sort last

2B. 2/Sort first

2C. 2/Sort last

Subject 10

Exemplify and explain the ray tracing algorithm for 3 levels of recursion and a scene consisting of:

- 1A.** 1 light source, 2 cubes (1 glass, 1 iron)
- 1B.** 2 light sources, 2 cubes (1 glass, 1 wood)
- 1C.** 1 light source, 2 cubes (1 glass, 1 chalk)
- 2A.** 2 light sources, 2 cubes (1 glass, 1 brick)
- 2B.** 1 light source, 2 cubes (1 iron, 1 chalk)
- 2C.** 2 light sources, 2 cubes (1 wood, 1 chalk)

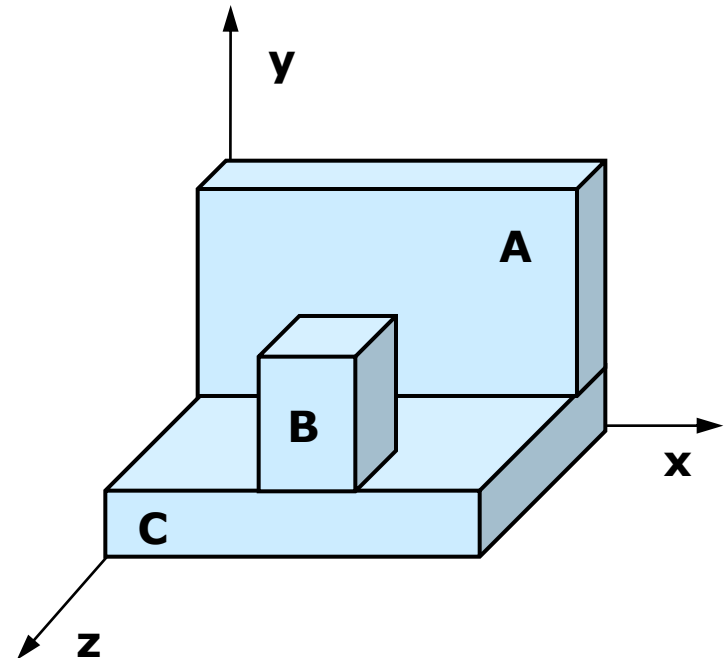
Graphical Processing Systems

Knowledge Assessment

Subject 1

Exemplify and explain for the complex object in the figure the following hidden removal algorithm. The viewer is on z axis and the projection is orthogonal along z:

- 1A.** Back-face culling
- 1B.** Area subdivision: Warnock
- 1C.** Depth buffer
- 2A.** Depth sorting
- 2B.** Scan-line
- 2C.** Area subdivision: Weiler-Atherton



Subject 2

Exemplify the visualization algorithm based on the BSP Tree for the following object:

1A.



1B.



1C.



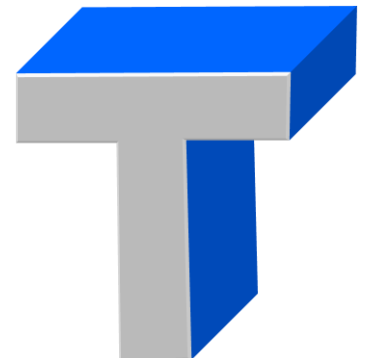
2A.



2B.



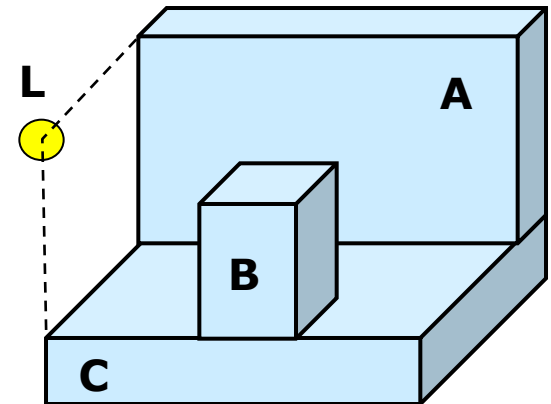
2C.



Subject 3

Exemplify and explain the following shadow computation algorithm for the complex object and the point light source in the next figure:

- 1A.** Shadows projected onto the ground plane
- 1B.** Scan line algorithm integrating the shadow polygons
- 1C.** Shadow volumes
- 2A.** Derivation of shadow polygons from light source transformations
- 2B.** Shadow Z-Buffer
- 2C.** Shadow volumes



Subject 4

Explain the 3D clipping and the perspective transformation from view space to screen space. Highlight and explain the main benefits and disadvantages of working in the 3D screen space.

Subject 5

Exemplify the computation of the light intensity by Gouraud and Phong interpolation for the given 2 polygons. Detail the Phong local reflection model. The light source is just a point at the specified position:

1A. Polygons: T, M

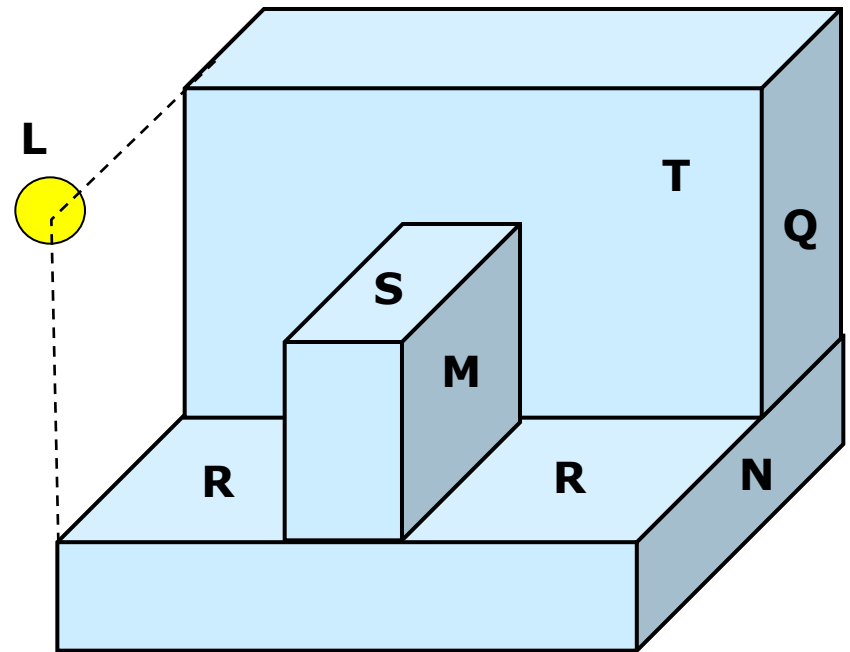
1B. Polygons: R, N

1C. Polygons: S, Q

2A. Polygons: R, Q

2B. Polygons: S, M

2C. Polygons: T, N



Subject 6

Exemplify the ray tracing algorithm for the points E and F in the specified polygons for 2 reflection levels. The user visualizes the objects on the screen as it is represented in the figure. The object B is transparent, and objects A and C opaque.

1A. E in T, F in M

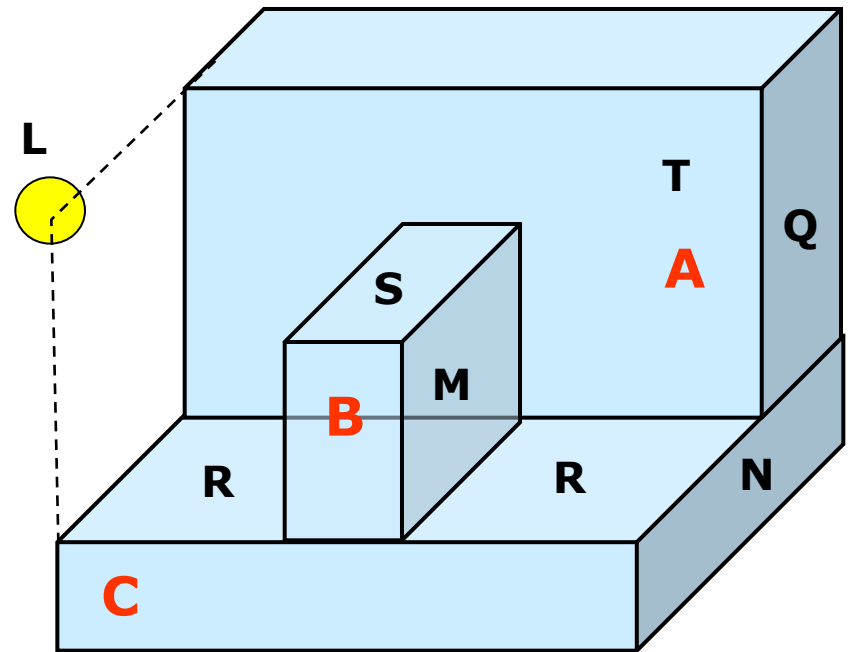
1B. E in R, F in N

1C. E in S, F in Q

2A. E in R, F in Q

2B. E in S, F in M

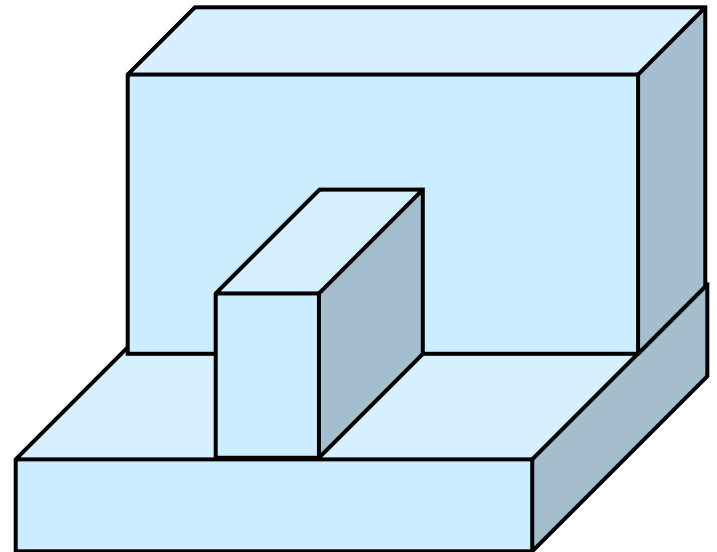
2C. E in T, F in N



Subject 7

Exemplify and explain for the following case and the object in the figure:

- 1A.** Texture mapping by Environment mapping technique
- 1B.** Texture mapping by two phases - S: T' is a sphere, O: surface normal
- 1C.** Rendering the CSG (Constructive Solid Geometry) object
- 2A.** Octree based computation of intersections in ray tracing
- 2B.** Bump mapping the texture on the polygonal surface
- 2C.** Texture mapping through modeling the 3D object



Subject 8

Explain a mathematical approach to:

- 1A.** Check if a polygon is concave
- 1B.** Check if a 3D polygonal object is convex
- 1C.** Compute the distance from a point to a plane
- 2A.** Check if two plane polygons intersect each other
- 2B.** The viewing ray intersects a 3D convex polygonal object
- 2C.** Two any cubes do not intersect each other

Subject 9

Explain and compare the classic and the GPU graphics pipeline.

Define and explain the Unified Shader Architecture

Subject 10

Explain a technique for morphic transformation of a cube into a pyramid.

Sisteme de Prelucrari Grafice

Examen

Subiect 1

Exemplificati si explicati pentru obiectul complex din figura, urmatorul algoritm pentru eliminarea liniilor si suprafetelor ascunse. Observatorul este pe axa z, iar proiectia este ortogonala de-a lungul axei z:

1A. Back-face culling

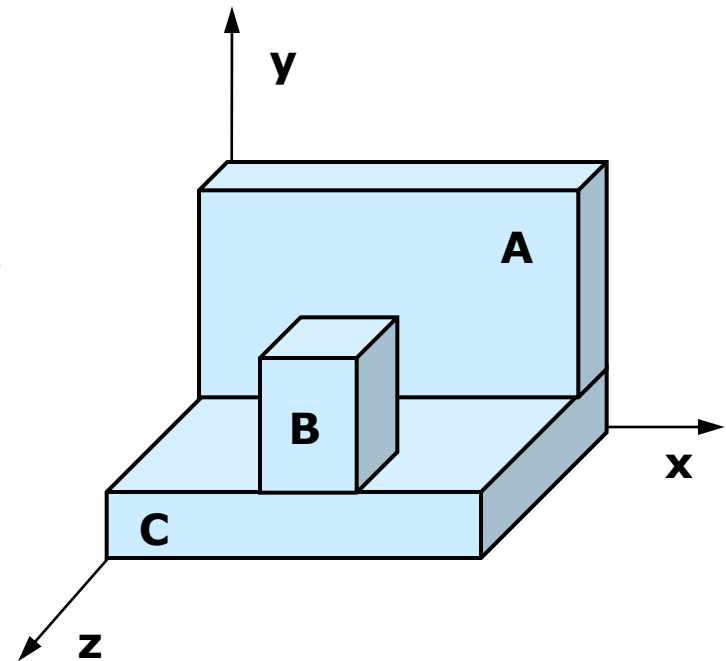
1B. Impartirea in arii elementare: Warnock

1C. Depth buffer

2A. Depth sorting

2B. Scan-line

2C. Impartirea in arii elementare: Weiler-Atherton



Subiect 2

Exemplificati vizualizarea urmatorului obiect 3D prin algoritmul BSP Tree:

1A.



1B.



1C.



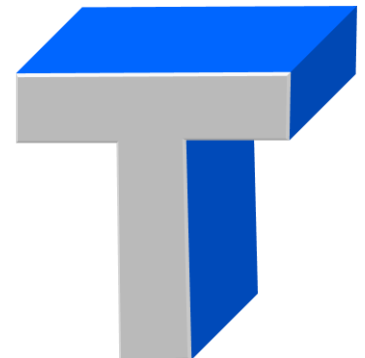
2A.



2B.



2C.



Subiect 3

Exemplificati si explicati urmatorul algoritm pentru calcularea umbrelor, pentru obiectul complex si sursa de lumina punctiforma din figura:

1A. Determinarea poligoanelor de umbra din transformari

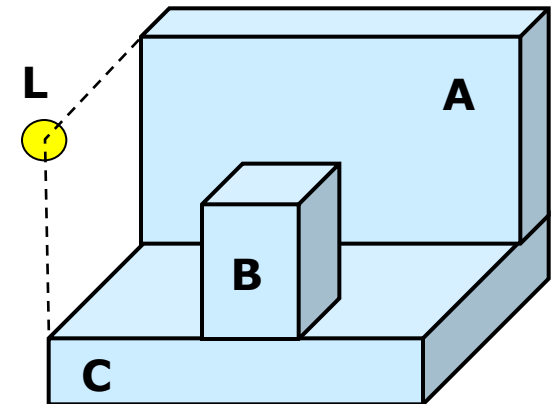
1B. Algoritmul scan line integrand poligoane de umbra

1C. Volume de umbra

2A. Umbre proiectate pe planul orizontal (podea)

2B. Z-Buffer de umbra

2C. Volume de umbra



Subiect 4

Explicati decuparea 3D si transformarea perspectiva din spatiul de vizualizare in spatiul ecran. Detaliati si explicati principalele avantaje si dezavantaje ale prelucrarilor grafice in spatiul ecran 3D.

Subject 5

Exemplificati calcularea intensitatii luminoase prin interpolarea Gouraud si Phong pentru doua poligoane date. Detaliati modelul de reflexie locala Phong. Sursa de lumina este punctiforma, plasata in pozitia specificata in figura:

1A. Poligoane: T, M

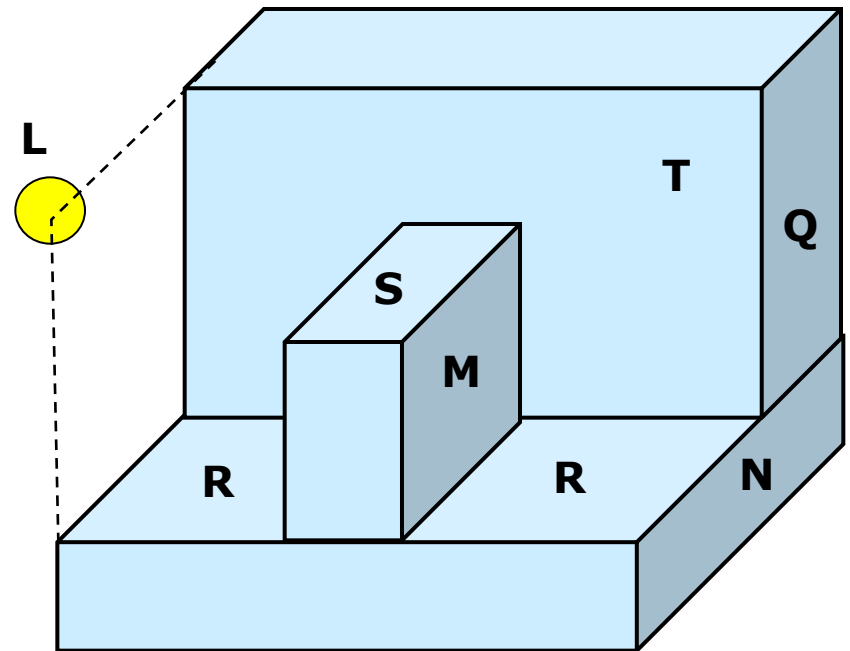
1B. Poligoane: R, N

1C. Poligoane: S, Q

2A. Poligoane: R, Q

2B. Poligoane: S, M

2C. Poligoane: T, N



Subject 6

Exemplificati algoritmul ray tracing pentru punctele E si F din poligoanele date, pana la nivelul 2 de reflexie. Utilizatorul vizualizeaza obiectele pe ecran asa cum sunt ele prezentate in figura. Obiectul B este transparent, iar obiectele A si C sunt opace.

1A. E in T, F in M

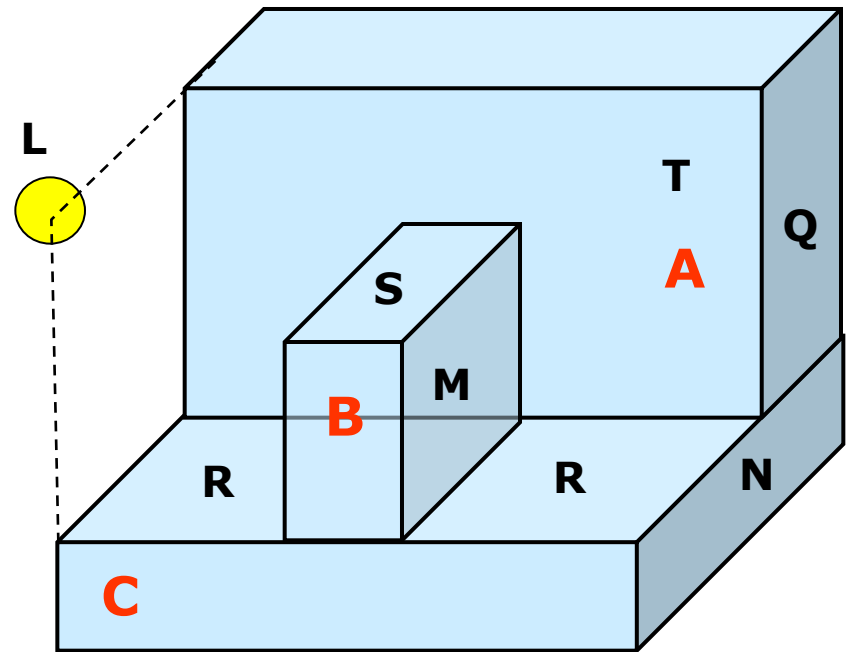
1B. E in R, F in N

1C. E in S, F in Q

2A. E in R, F in Q

2B. E in S, F in M

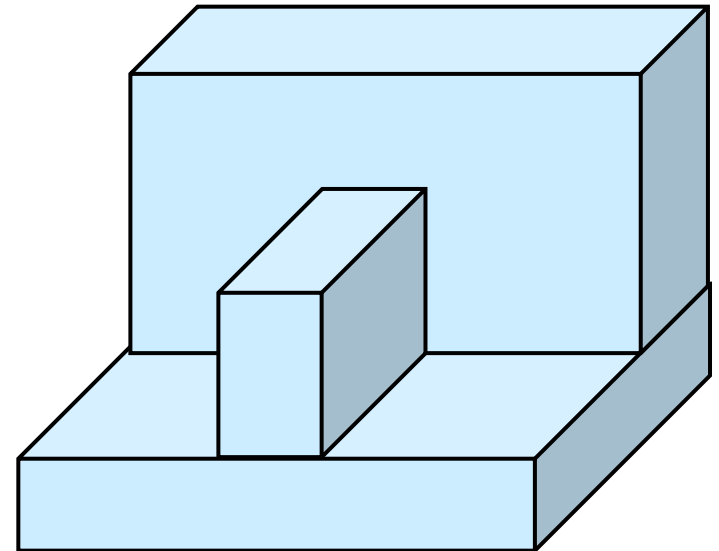
2C. E in T, F in N



Subiect 7

Exemplificati si explicati pentru urmatorul caz si obiect din figura:

- 1A.** Maparea texturii prin tehnica maparea mediului (Environment mapping)
- 1B.** Maparea texturii prin 2 faze - S: T' este o sfera, O: normala la suprafata intermediara
- 1C.** Trasarea unui obiect CSG (Constructive Solid Geometry)
- 2A.** Calcularea intersectiilor in algoritmul ray tracing folosind modelarea Octree
- 2B.** Bump mapping pe suprafata 3D modelata prin poligoane
- 2C.** Maparea texturii prin modelarea obiectului 3D



Subiect 8

Explicati o metoda matematica pentru a determina:

- 1A.** Daca un poligon dat este concav
- 1B.** Daca un obiect 3D este convex
- 1C.** Distanta de la un punct dat la o piramida triunghiulara
- 2A.** Daca doua poligoane plane se intersecteaza
- 2B.** Daca raza vizuala intersecteaza o piramida triunghiulara
- 2C.** Daca doua piramide triunghiulare oarecare se intersecteaza

Subject 9

Explicati si comparati secventa de transformari grafice (pipeline) clasica si GPU. Definiti si explicati arhitectura GPU numita Unified Shader Architecture

Subiect 10

Explicati o tehnica pentru transformarea morfologica a unei piramide triunghiulare intr-un cub (morfism).

Graphical Processing Systems

Examination Subjects

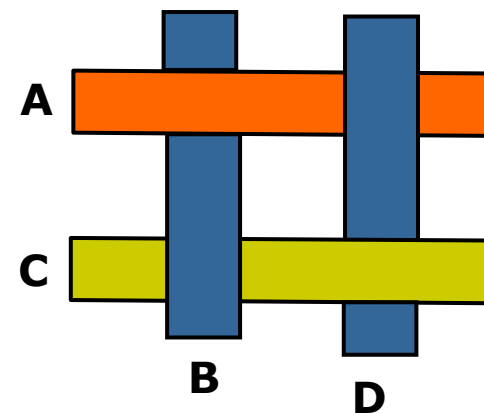
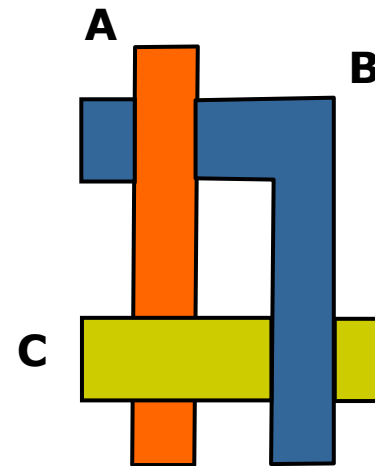
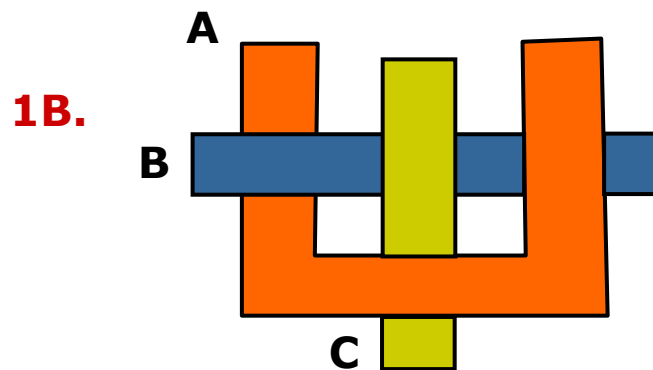
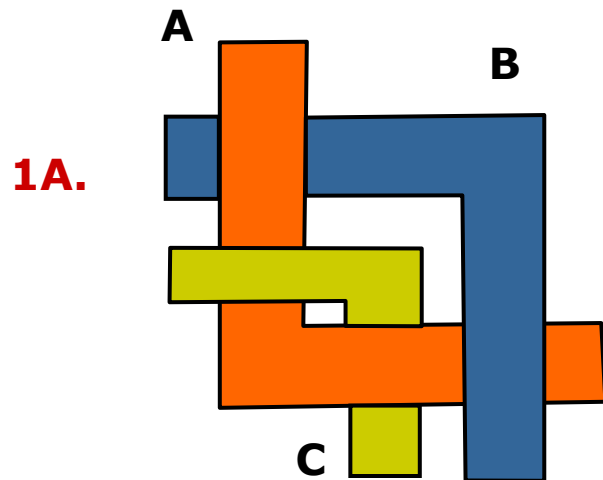
Subject 1

Exemplify and explain an approach to compute efficiently the first intersection between a viewing ray and a set of three objects, if they are:

- 1A.** One cube and two spheres.
- 1B.** One cone, one cube and one sphere.
- 2A.** One sphere, one cone and one cylinder.
- 2B.** Two cones and one cylinder.

Subject 2

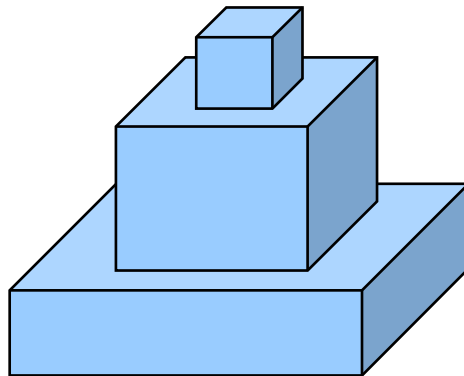
Explain the Z-Buffer algorithm for hidden surface removal in the case of the following set of objects:



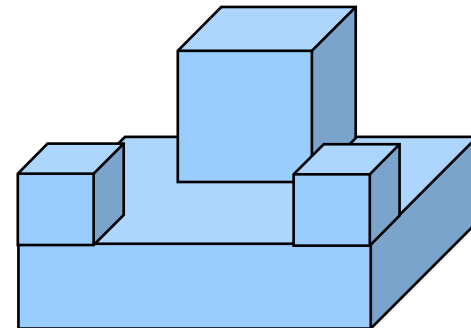
Subject 3

Exemplify the BSP Tree for the following object. Explain the parsing of the tree at the scene visualization.

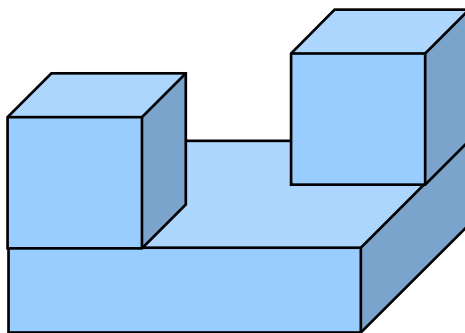
1A.



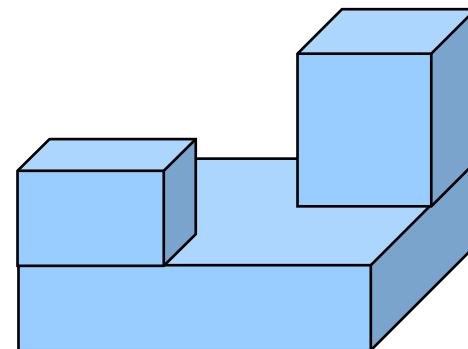
2A.



1B.



2B.



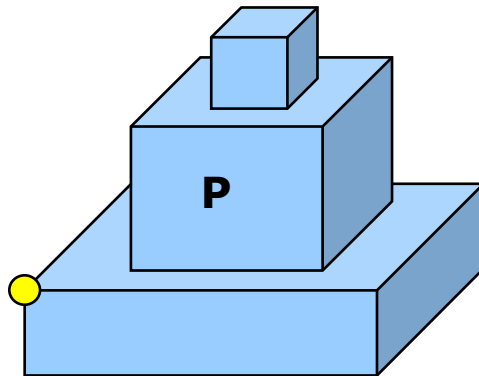
Subject 4

Explain the 3D clipping and the perspective transformation from view space to screen space. What are the main advantages and disadvantages of working in the screen space.

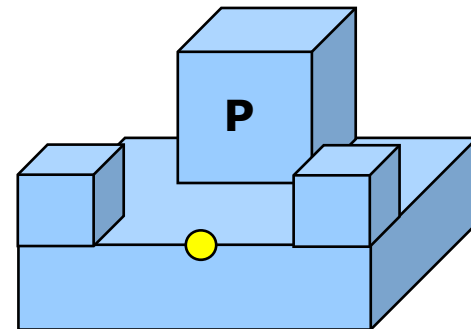
Subject 5

Exemplify the computation of the light intensity by Phong local reflection model for the following objects. Detail the Gouraud and Phong shading for the polygon P, if the light source is just a point in the specified position:

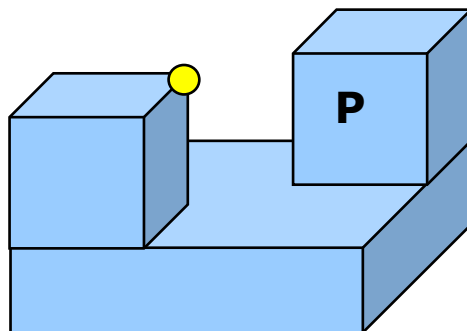
1A.



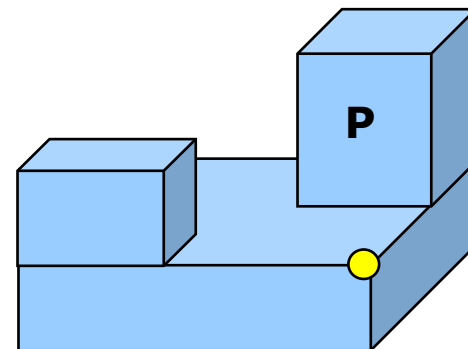
2A.



1B.



2B.



Subject 6

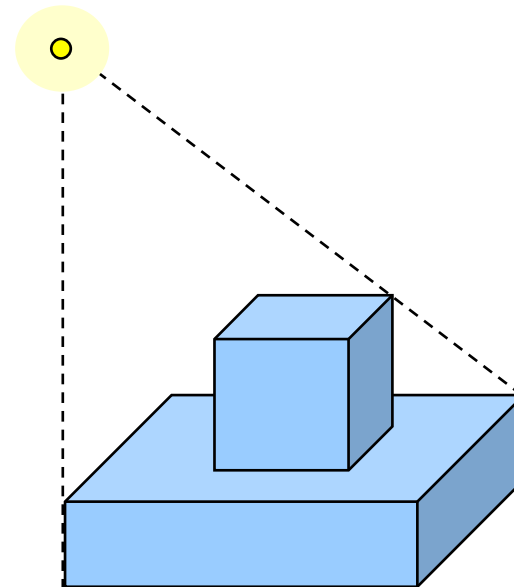
Exemplify and explain the computation of shadows by the following algorithm and the objects:

1A. Shadow Z-Buffer

1B. Shadow volumes

2A. Transformations

2B. Shadow scanline



Subject 7

Exemplify and explain the two phase inverse texture mapping technique for the following case, if the object is a cube:

- 1A.** S: T' is a sphere, O: surface normal
- 1B.** S: T' is a cylinder, O: object's normal
- 2A.** S: T' is a cube, O: reflected view
- 2B.** S: T' is a cylinder, O: reflected view

Subject 8

Exemplify and explain the ray tracing algorithm on the objects given at the subject 1.

Subject 9

Explain and compare the classic and the GPU graphics pipeline.

Define and explain the Unified Shader Architecture

Subject 10

Explain and compare the sort first and sort last graphics pipeline on the objects given at the subject 5.

Sisteme de Prelucrari Grafice

Subiecte pentru examen

Subiect 1

Exemplificati si explicati o metoda pentru calcularea eficienta a primei intersectii dintre raza vizuala si o scena formata din 5 obiecte:

1A. 3 cuburi si 2 sfere.

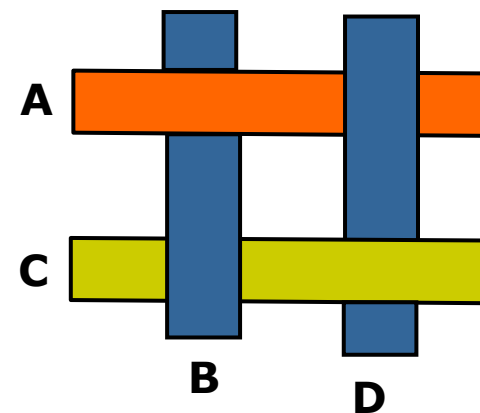
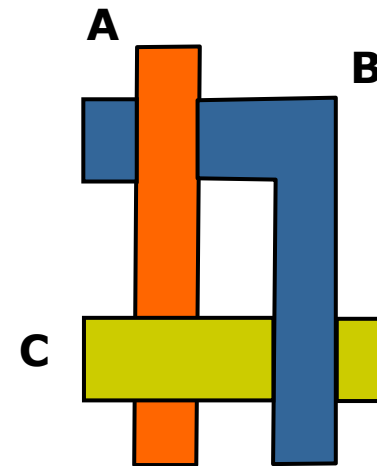
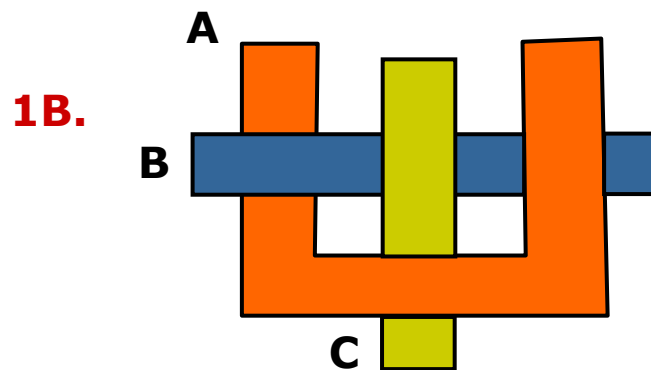
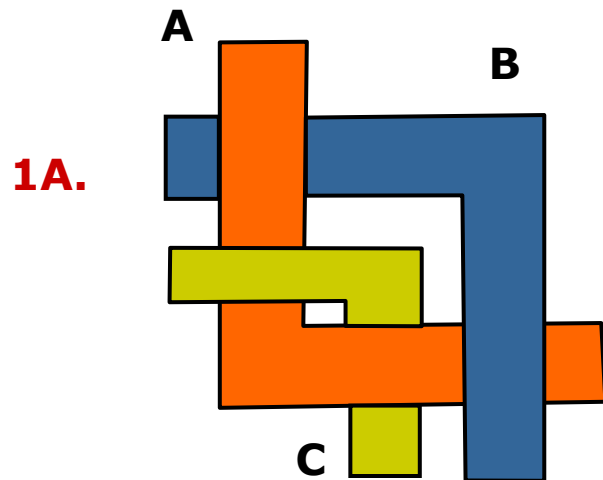
1B. 2 cilindri, 1 cub si 2 sfere.

2A. 2 sfere si 3 cilindri.

2B. 3 cuburi si 2 cilindri.

Subiect 2

Explicati decuparea in planul de proiectie a urmatoarelor poligoane prin algoritmul Weiler:

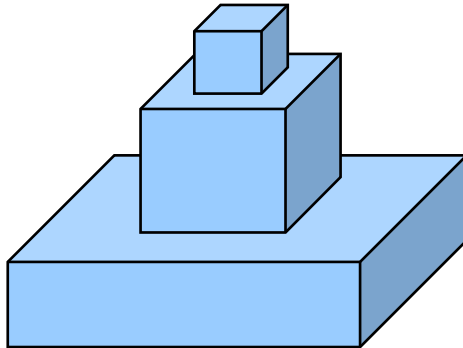


Subiect 3

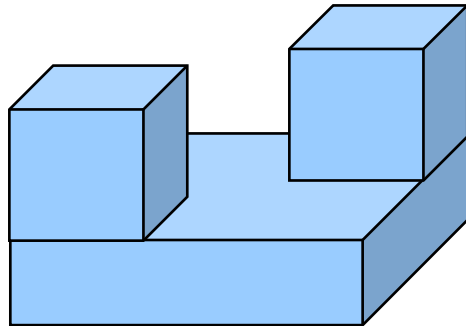
Exemplificati modelul Octree pentru obiectele date in urmatoarea figura.

Observatie: Se impune forma si pozitia relativa a obiectelor. Este la libera alegere dimensiunea lor.

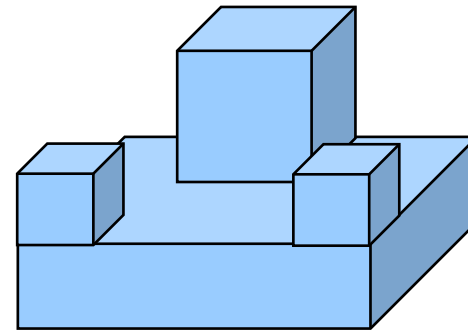
1A.



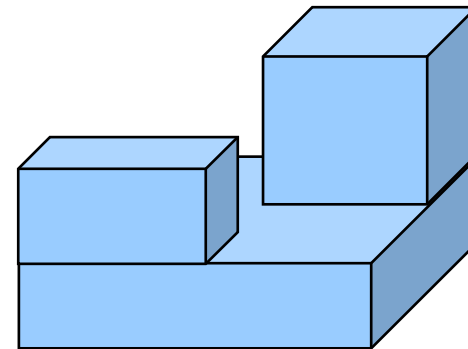
1B.



2A.



2B.

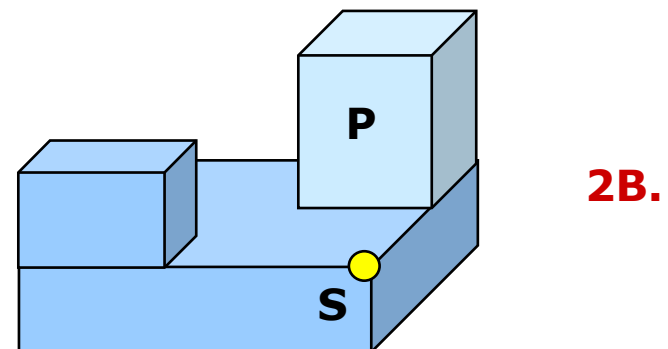
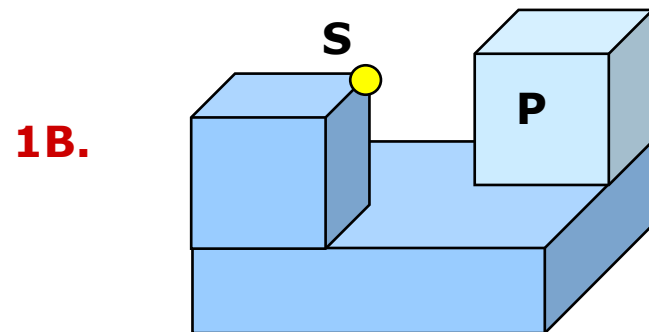
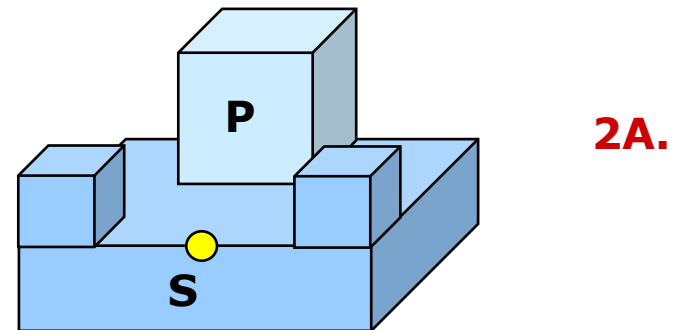
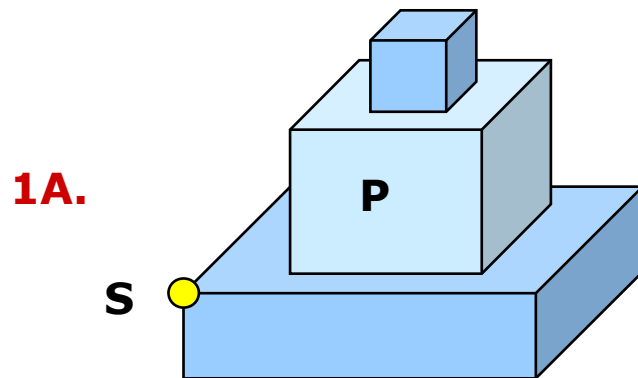


Subiect 4

Explicati secventa de transformare a unui obiect 3D din spatiul de vizualizare in spatiul ecran. Care sunt avantajele si dezavantajele acestei transformari.

Subiect 5

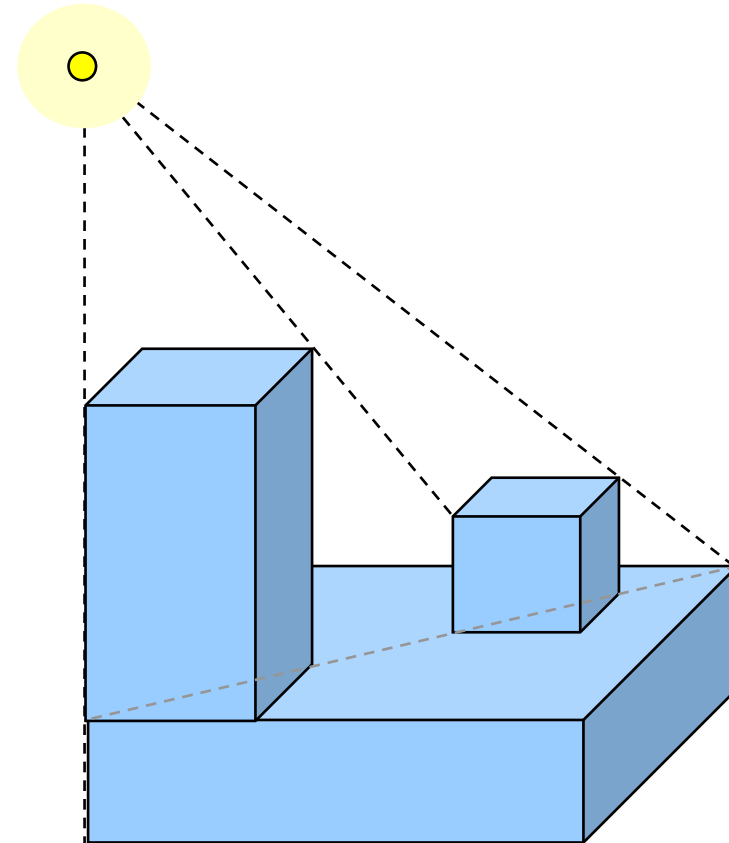
Explicati calcularea intensitatii luminoase folosind modelul de reflexie locala Phong, pentru urmatoarele obiecte. Detaliati calcularea incrementala a intensitatii luminoase prin metodele Gouraud si Phong pentru poligonul P, daca sursa de lumina punctiforma este in pozitia S data in figura:



Subiect 6

Exemplificati si explicati calcularea umbrelor pentru obiectele din figura prin urmatorul algoritm :

- 1A.** Z-Buffer de umbra
(Shadow Z-Buffer)
- 1B.** Volume de umbra
(Shadow volumes)
- 2A.** Transformari
- 2B.** Scanline cu umbre
(Shadow scanline)



Subiect 7

Exemplificati si explicati tehnica inversa de mapare a unei texturi prin doua faze, pentru urmatorul caza, daca obiectul este un cilindru:

1A. S: T' este o sfera, O: normala la suprafata

1B. S: T' este un cub, O: normala la obiect

2A. S: T' este un cub, O: raza vizuala

2B. S: T' este un cilindru, O: raza vizuala

Subiect 8

Exemplificati si explicati algoritmul ray tracing pentru obiectele date la Subiectul 1 si doua surse de lumina. Primele 3 obiecte din lista sunt transparente, iar ultimele 2 sunt opace.

Subiect 9

Explicati si comparati transformarile dintr-un pipeline grafic clasic cu cele ale unui GPU. Definiti si explicati Arhitcetura Shader Unificat (Unified Shader Architecture).

Subiect 10

Explicati si comparati secventele grafice pipeline "sort first" si "sort pentru obiectele date la Subiectul 2, daca in final se obtine aceeaasi imagine ca in figura.

Graphical Processing Systems

Examination Subjects

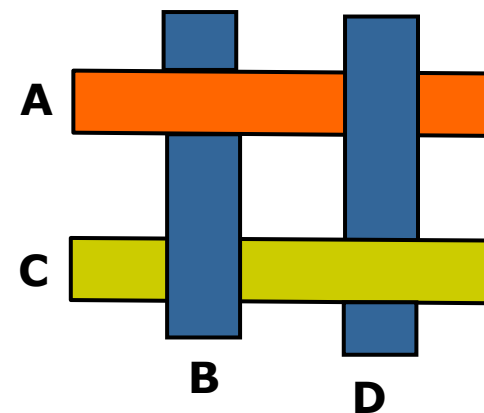
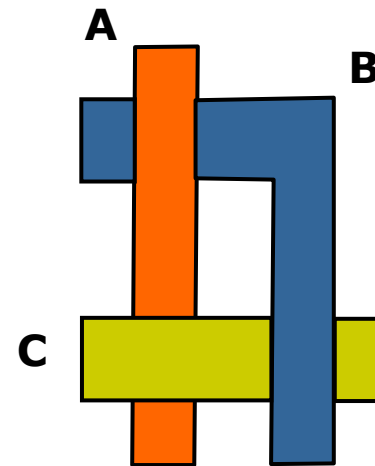
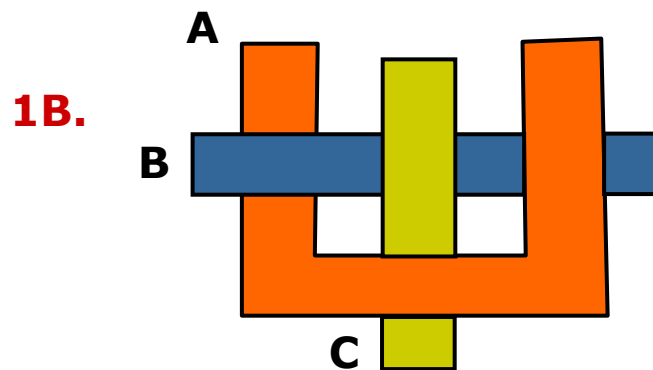
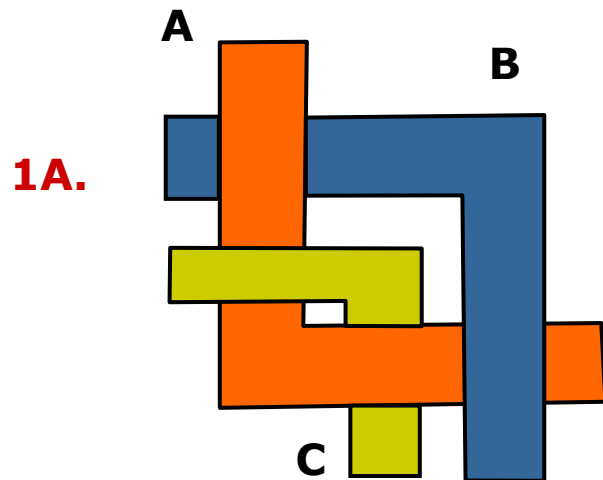
Subject 1

Exemplify and explain an approach to compute efficiently the first intersection between a viewing ray and a set of three objects, if they are:

- 1A.** One cube and two spheres.
- 1B.** One cone, one cube and one sphere.
- 2A.** One sphere, one cone and one cylinder.
- 2B.** Two cones and one cylinder.

Subject 2

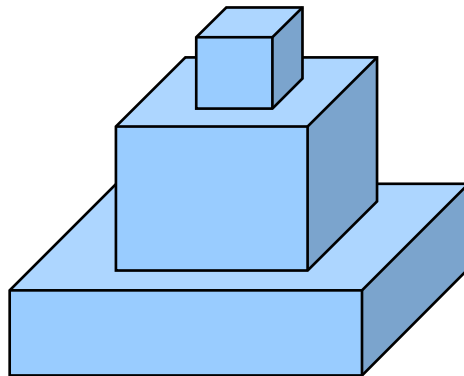
Explain the Z-Buffer algorithm for hidden surface removal in the case of the following set of objects:



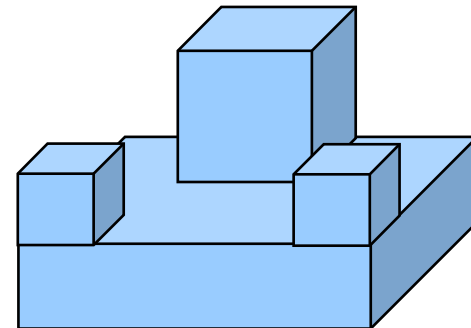
Subject 3

Exemplify the BSP Tree for the following object. Explain the parsing of the tree at the scene visualization.

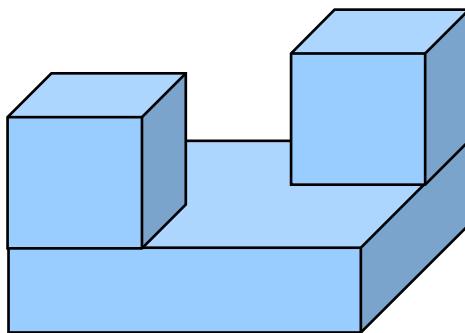
1A.



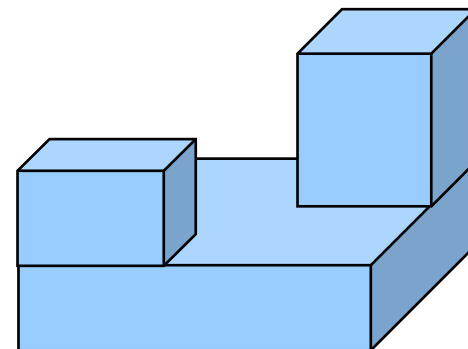
2A.



1B.



2B.



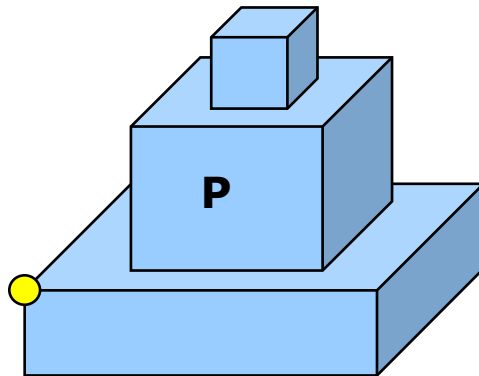
Subject 4

Explain the 3D clipping and the perspective transformation from view space to screen space. What are the main advantages and disadvantages of working in the screen space.

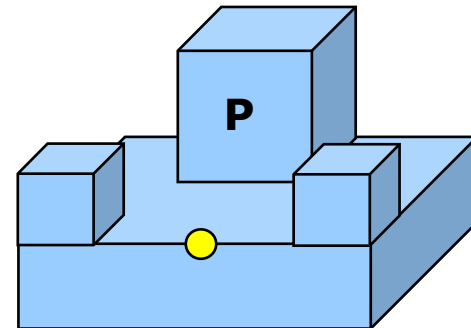
Subject 5

Exemplify the computation of the light intensity by Phong local reflection model for the following objects. Detail the Gouraud and Phong shading for the polygon P, if the light source is just a point in the specified position:

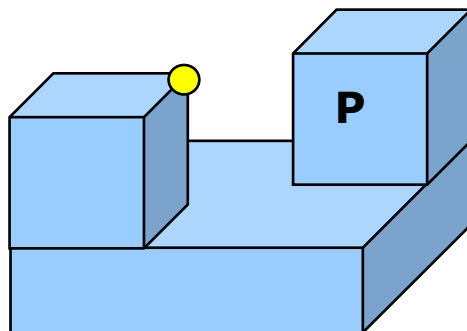
1A.



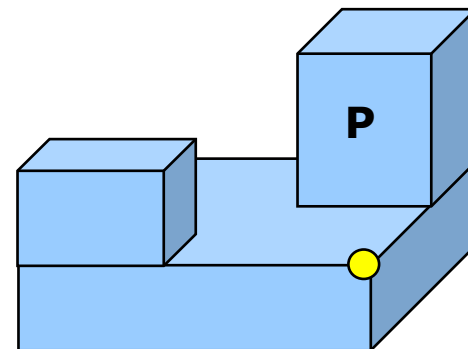
2A.



1B.



2B.



Subject 6

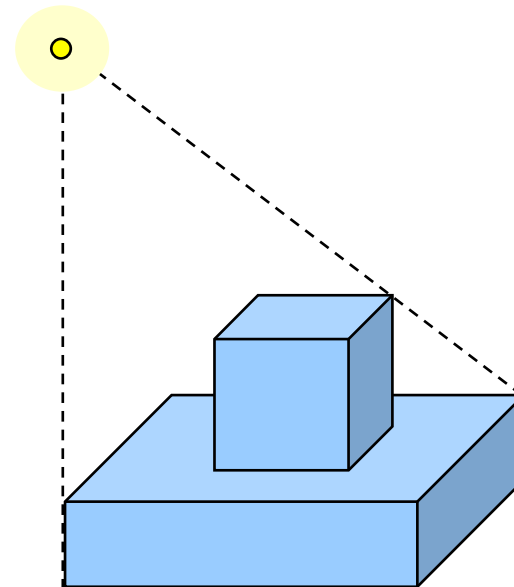
Exemplify and explain the computation of shadows by the following algorithm and the objects:

1A. Shadow Z-Buffer

1B. Shadow volumes

2A. Transformations

2B. Shadow scanline



Subject 7

Exemplify and explain the two phase inverse texture mapping technique for the following case, if the object is a cube:

- 1A.** S: T' is a sphere, O: surface normal
- 1B.** S: T' is a cylinder, O: object's normal
- 2A.** S: T' is a cube, O: reflected view
- 2B.** S: T' is a cylinder, O: reflected view

Subject 8

Exemplify and explain the ray tracing algorithm on the objects given at the subject 1.

Subject 9

Explain and compare the classic and the GPU graphics pipeline.

Define and explain the Unified Shader Architecture

Subject 10

Explain and compare the sort first and sort last graphics pipeline on the objects given at the subject 5.

Graphical Processing Systems

Examination Subjects

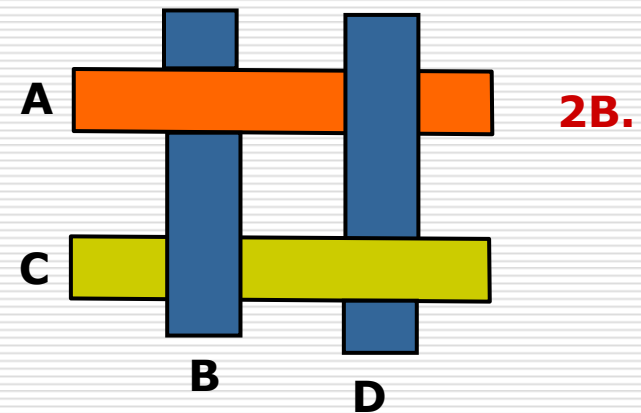
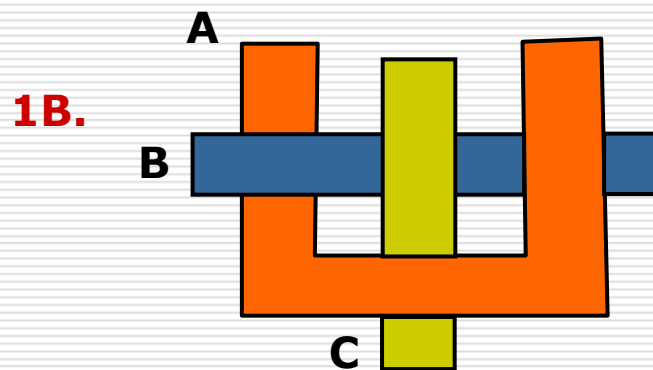
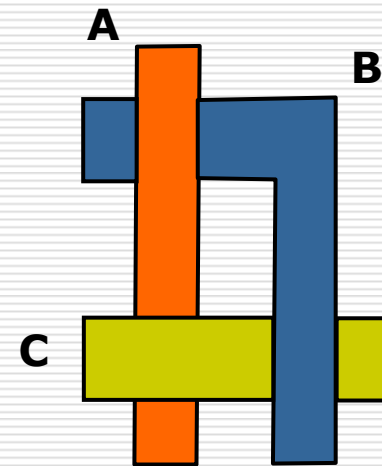
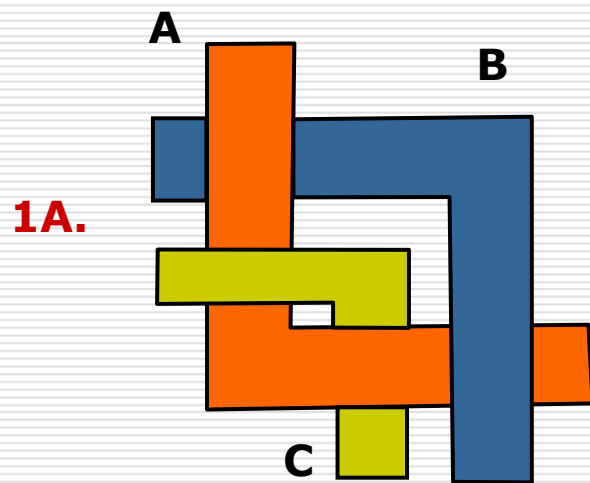
Subject 1

Exemplify and explain a mathematical approach to compute the case:

- 1A.** Polygon triangularization.
- 1B.** Minimum distance from a point P to a cube.
- 2A.** A polygonal 3D object is convex.
- 2B.** Relative position of two triangles: disjoint, included, surrounded, intersected

Subject 2

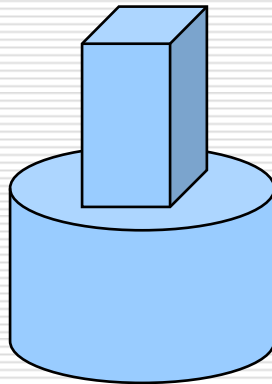
Explain the Weiler–Atherton algorithm for hidden surface removal



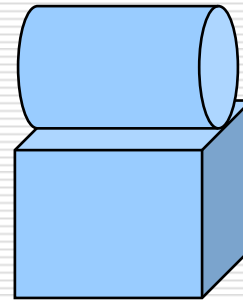
Subject 3

Exemplify the BSP Tree for the following object. Explain the tree parsing at the scene visualization.

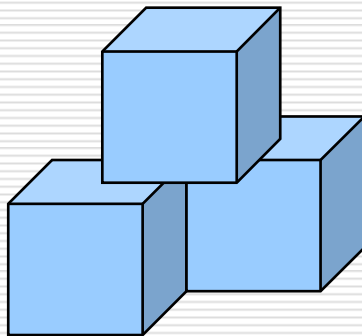
1A.



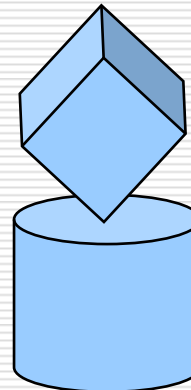
2A.



1B.



2B.



Subject 4

View volume definition. Perspective transformation from view space to screen space. Advantages and disadvantages.

Subject 5

Phong local reflection model. Color computation.

Explain how can be modeled an object as:

1A. Metallic of green color.

1B. Matte of yellow color.

2A. Bright sphere reflecting two large light sources. What is the color of the light patches on the sphere?

2B. Polished object of brown wood.

Subject 6

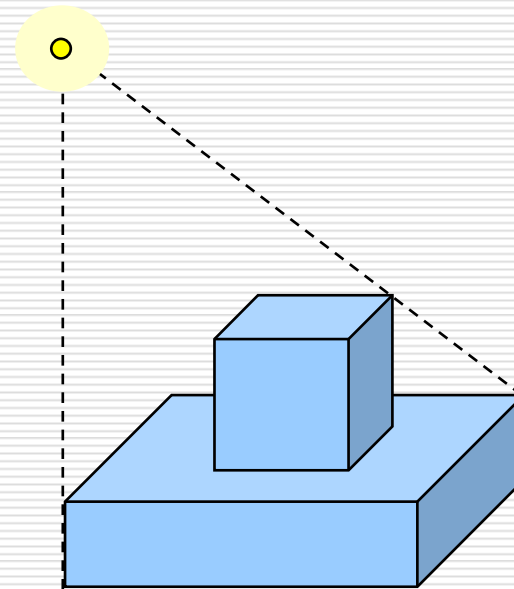
Exemplify and explain the computation of shadows by the following algorithm and the object:

1A. Shadow Z-Buffer

1B. Shadow volumes

2A. Transformations

2B. Shadow scanline



Subject 7

Exemplify and explain the two phase inverse texture mapping technique for the following case, if the object is a cube:

- 1A.** S: T' is a sphere, O: surface normal
- 1B.** S: T' is a cylinder, O: object's normal
- 2A.** S: T' is a cube, O: reflected view
- 2B.** S: T' is a cylinder, O: reflected view

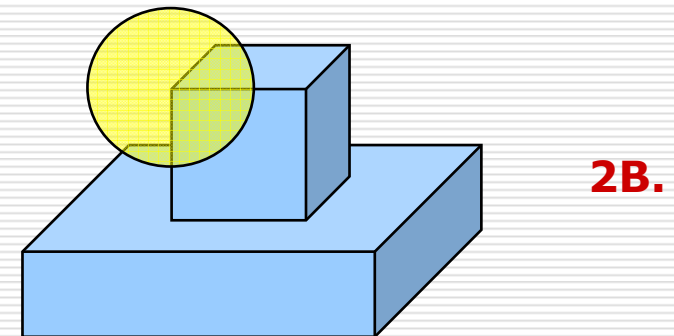
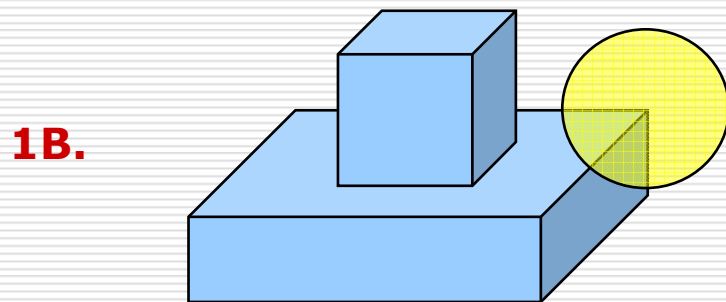
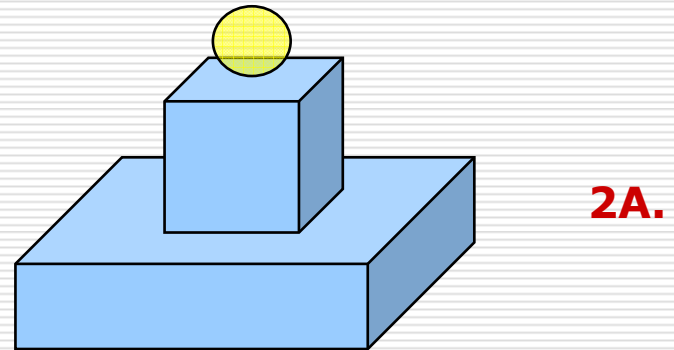
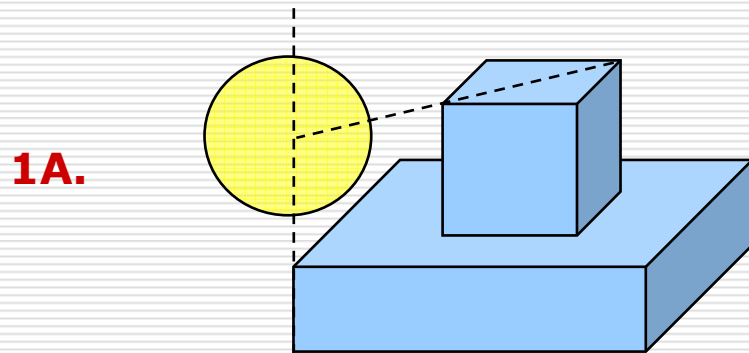
Subject 8

Ray tracing

- 1A.** Explain the basic and the recursive ray tracing algorithm
- 1B.** Global reflection model
- 2A.** Spatial coherence. Octree based ray tracing computation
- 2B.** Spatial coherence. SEADS based ray tracing computation

Subject 9

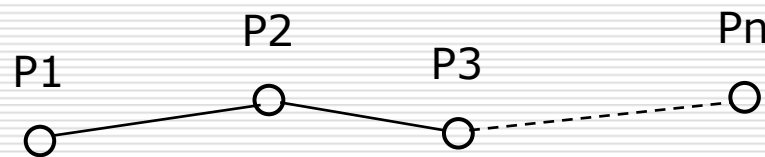
Exemplify the radiosity algorithm for the following scene of objects:



Subject 10

Animation. Exemplify and explain the movement, speed control, and time control in object animation along:

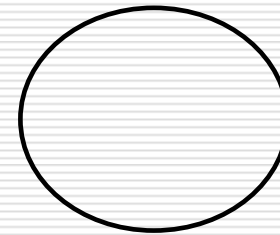
1A. Polyline $P1..Pn$



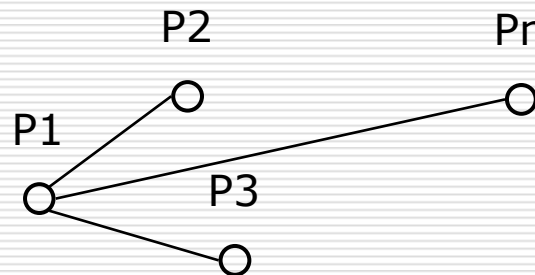
1B. Square



2A. Circle



2B. Linear network



Sisteme de Prelucrari Grafice

Subiecte Examen

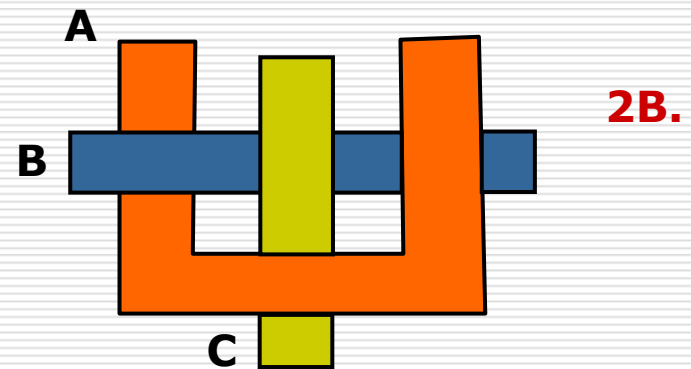
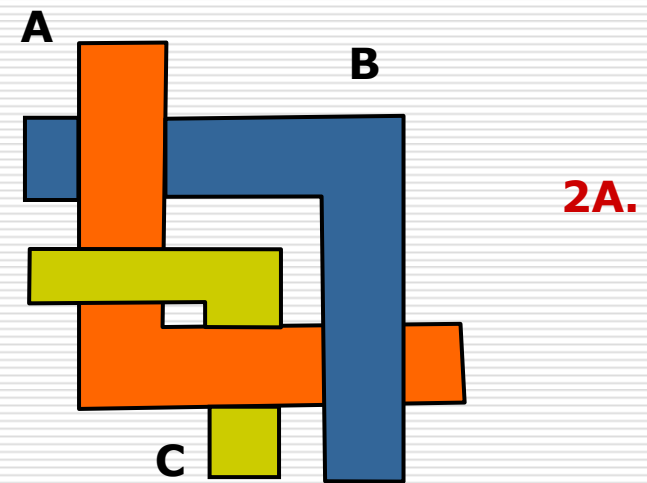
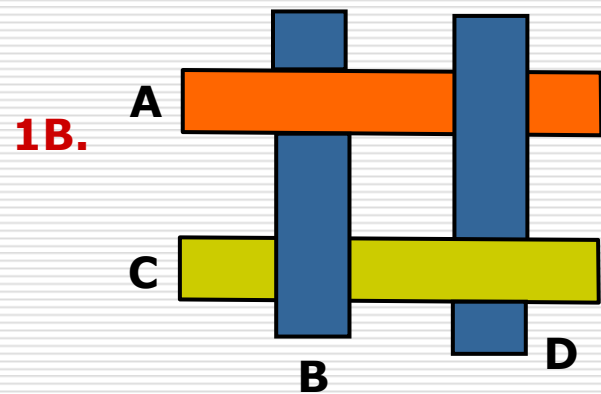
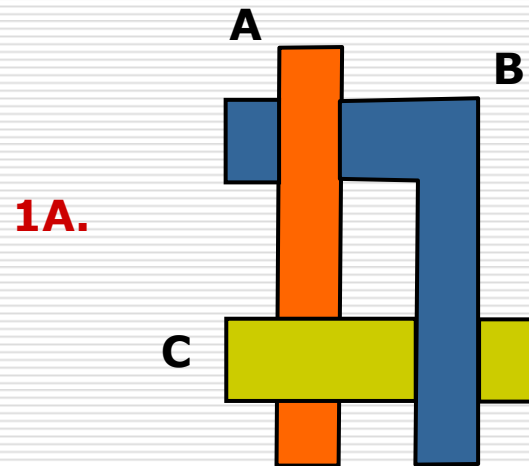
Subiect 1

Exemplificati si explicati o metoda matematica pentru calcularea:

- 1A.** Distanta minima de la un punct P la un cub
- 1B.** Triangularizarea unui poligon
- 2A.** Pozitia relativa a doua triunghiuri: disjuncte, inclus, inconjurator, intersectate
- 2B.** Determinarea cazului daca un obiect poligonal 3D este convex

Subiect 2

Explicati algoritmul Weiler–Atherton pentru eliminarea muchiilor si fetelor ascunse pentru urmatoarele poligoane:



Subiect 4

Definitia volumului de vizualizare. Transformarea perspectiva din spatiul de vizualizare in spatiul ecran. Avantaje si dezavantaje.

Subiect 5

Modelul de reflexie locala Phong. Calcularea culorii. Explicati cum poate fi modelat urmatorul obiect:

- 1A.** Obiect slefuit din lemn maro
- 1B.** Sfera lucioasa reflectand doua surse mari de lumina. Ce culoare au petele de lumina de pe sfera?
- 2A.** Mat de culoare galbena
- 2B.** Metalic de culoare verde

Subiect 6

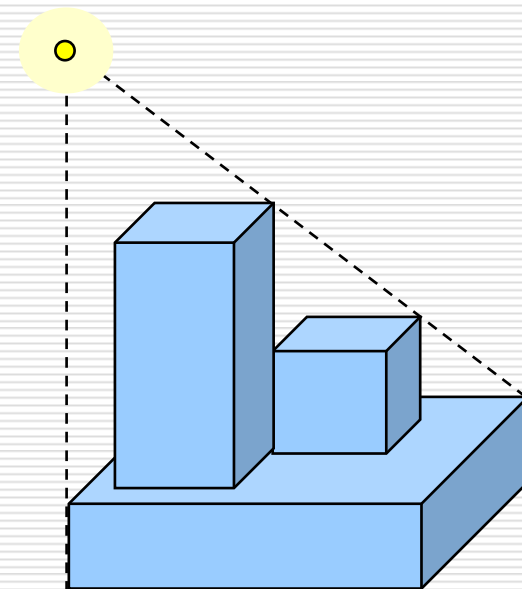
Exemplificati si explicati calcularea umbrelor pentru urmatorul algoritm si obiect:

1A. Z-Buffer de umbre (Shadow Z-Buffer)

1B. Volum de umbra (Shadow volume)

2A. Transformari

2B. Linie de baleiaj (Shadow scanline)



Subiect 7

Exemplificati si explicati maparea inversa a unei texturi in doua faze, pentru urmatorul caz, daca obiectul este un cub:

1A. S: T' este un cilindru, O: normala la suprafata obiectului

1B. S: T' este o sfera, O: normala la suprafata intermediara

2A. S: T' este un cilindru, O: reza vizuala reflectata

2B. S: T' este un cub, O: raza vizuala reflectata

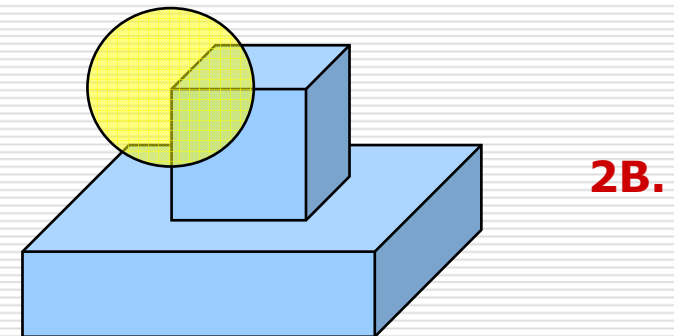
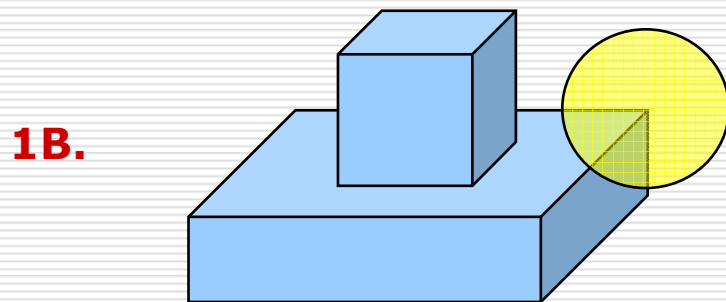
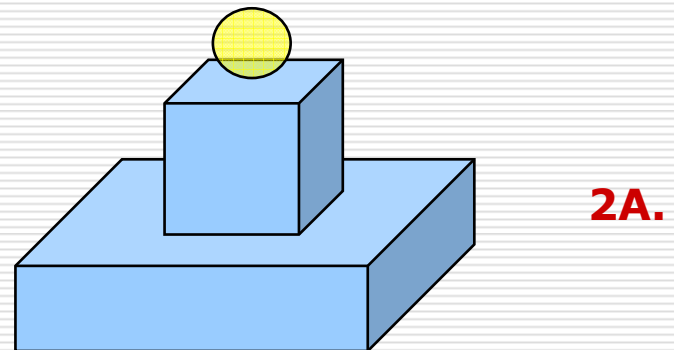
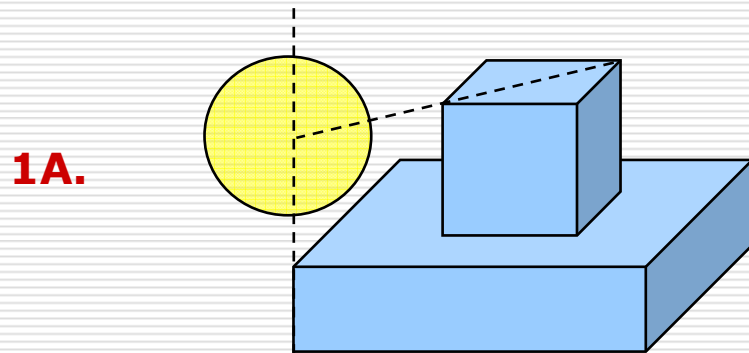
Subiect 8

Ray tracing

- 1A.** Coerenta spatiala. Trasarea bazata pe descrierea octree
- 1B.** Modelul de reflexie globala
- 2A.** Explicati principiul algoritmului ray tracing de baza si recursiv
- 2B.** Coerenta spatiala. Trasarea bazata pe descrierea SEADS

Subiect 9

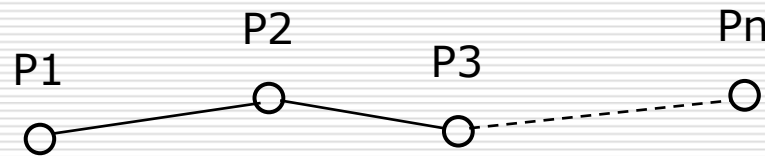
Exemplificati algoritmul radiatiei (Radiosity) pe urmatoarea scena de obiecte:



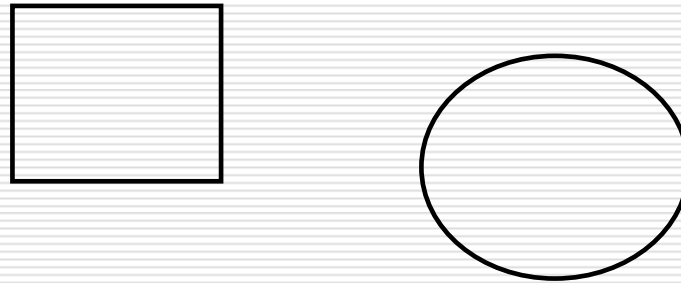
Subiect 10

Animatie. Exemplificati si explicati deplasarea, controlul vitezei si, controlul timpului pentru animatia unui obiect de-a lungul unei traiectorii de forma:

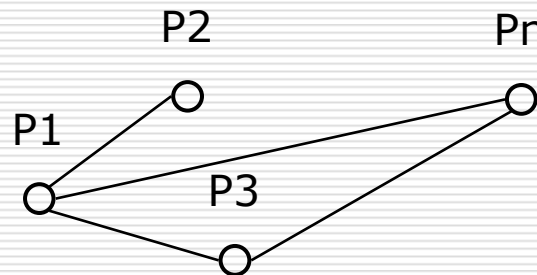
1A. Polilinie $P1..Pn$



1B. Cerc



2A. Patrat



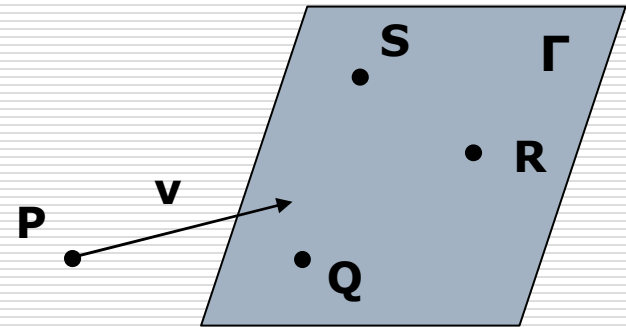
2B. Retea de linii

Sisteme de Prelucrari Grafice

Subiecte Examen

Subiect 1

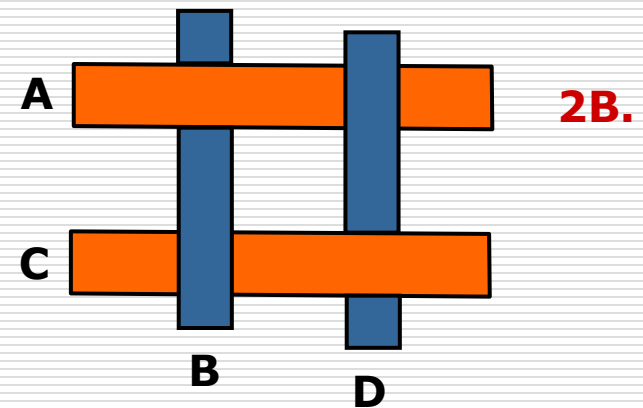
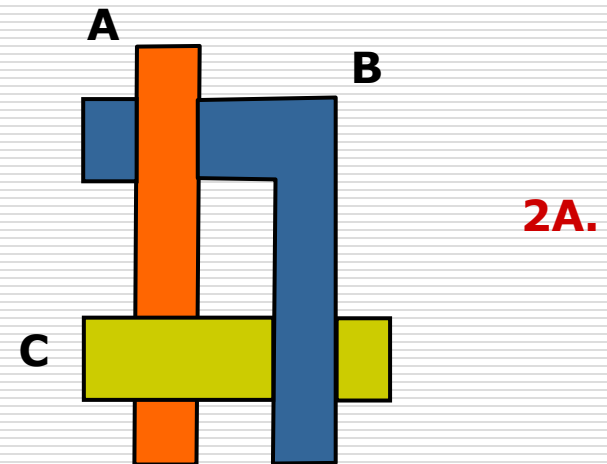
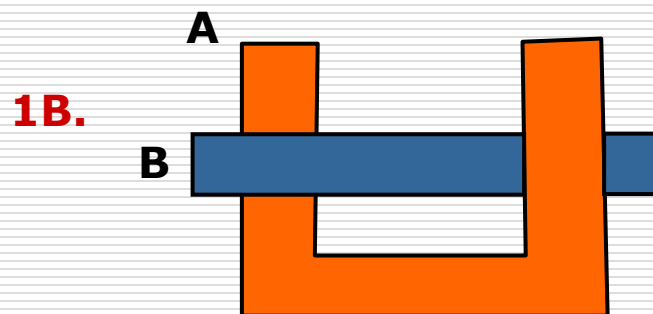
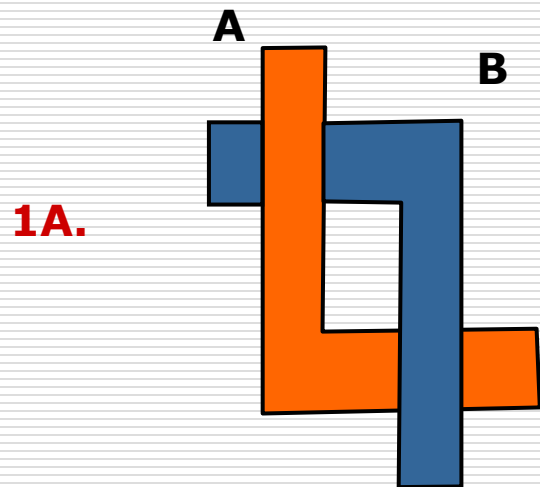
Fie un punct P exterior unui plan Γ definit de trei puncte Q , R și S .



- 1A.** Sa se determine pozitia punctului P in raport cu planul Γ : in fata, in spate, in plan.
- 1B.** Sa se calculeze distanta de la punctul P la planul Γ .
- 2A.** Daca un observator priveste planul Γ din pozitia P in directia v , sa se calculeze intersectia dintre raza vizuala si plan.
- 2B.** Sa se calculeze proiectia M a punctului P pe planul Γ si sa se determine daca M este in interiorul triunghiului QRS .

Subiect 2

Explicati algoritmul Weiler–Atherton pentru urmatorul exemplu:



Subiect 3

- 1A.** Secventa de trasare a unui obiect poligonal (rendering pipeline).
Descrierea generala.
- 1B.** Trasarea unui obiect definit parametric (parametric patch net).
- 2A.** Trasarea obiectelor definite CSG (CSG description).
- 2B.** Transformarea din spatiul de vizualizare in spatiul ecran.
Coordonate omogene.

Subiect 4

- 1A.** Tehnici incrementale de umbrire: interpolarea Gouraud.
- 1B.** Tehnici incrementale de umbrire: interpolarea Phong.
- 2A.** Componenta de reflexie difuza in modelul de reflexie locala.
- 2B.** Componenta de reflexie speculara in modelul de reflexie locala.

Subiect 5

Modelul de reflexie locala Phong. Implementare. Calcularea culorii.

Sa se explice cum se poate modela o suprafata:

- 1A.** Metalica, de culoare preponderent verde.
- 1B.** Mata, de culoare preponderent galben.
- 2A.** Sfera lucioasa cu reflexia a doua surse de lumina mari. Ce culoare are pata de lumina de pe sfera?
- 2B.** De intensitate luminoasa uniforma fara caracteristici preponderent mate sau lucioase.

Subiect 6

Calcularea formelor de umbra prin algoritmul:

1A. Scanline.

1B. Volume de umbra.

2A. Transformari.

2B. Z-Buffer de umbra.

Subiect 7

Maparea texturilor

- 1A.** Maparea lineara intr-un poligon plan.
- 1B.** Maparea cilindrica si sferica.
- 2A.** Maparea in doua faze.
- 2B.** Maparea inversa in doua faze. Exemplu – maparea S este pe o sfera, maparea O – normala la suprafata intermediara.

Subiect 8

Ray tracing

- 1A.** Explicati principiul algoritmului ray tracing de baza si recursiv.
- 1B.** Modelul de reflexie globala.
- 2A.** Coerenta spatiala. Trasarea bazata pe descrierea octree.
- 2B.** Coerenta spatiala. Trasarea bazata pe aborele BSP.

Subiect 9

- 1A.** Comparati modelele Phong, Ray tracing si Radiosity.
- 1B.** Exemplificati algoritmul radiatiei (Radiosity) pe figura 1B.
- 2A.** Exemplificati algoritmul radiatiei (Radiosity) pe figura 2A.
- 2B.** Exemplificati algoritmul radiatiei (Radiosity) pe figura 2B.

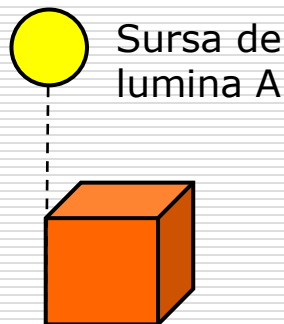


Fig. 1B.

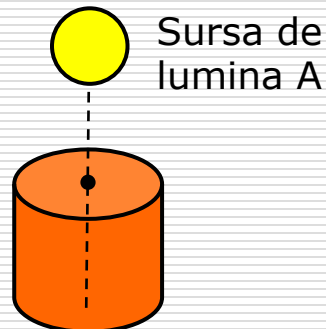


Fig. 2A.

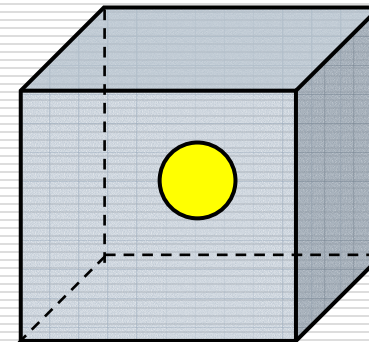


Fig. 2B.

Subiect 10

Animatia

- 1A.** Animatia prin modificarea caracteristicilor.
- 1B.** Animatia prin cadre cheie (key frames). Interpolarea liniara.
- 2A.** Controlul vitezei in interpolarea liniara.
- 2B.** Transformarea morfologica (morfism).

Sisteme de Prelucrari Grafice

Subiecte pentru examen

Subiect 1

Exemplificati si explicati o metoda pentru calcularea eficienta a primei intersectii dintre raza vizuala si o scena formata din 5 obiecte:

1A. 3 cuburi si 2 sfere.

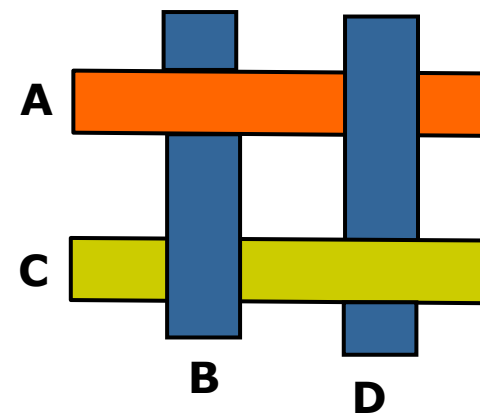
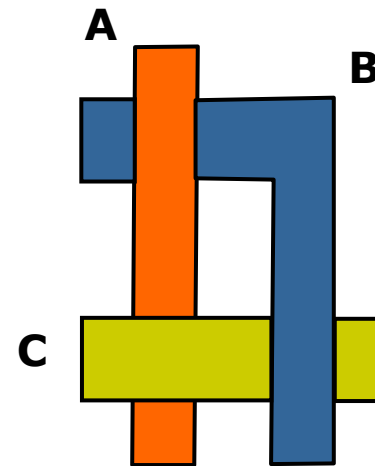
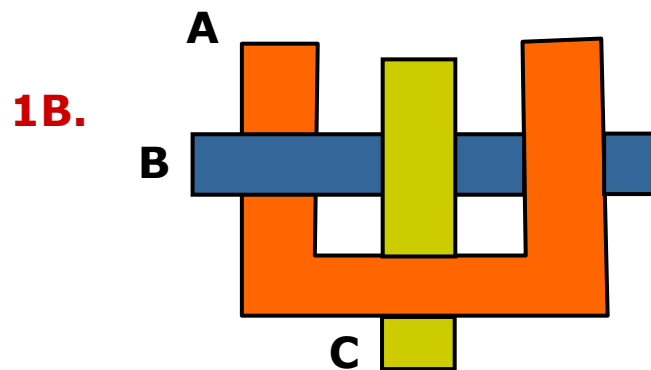
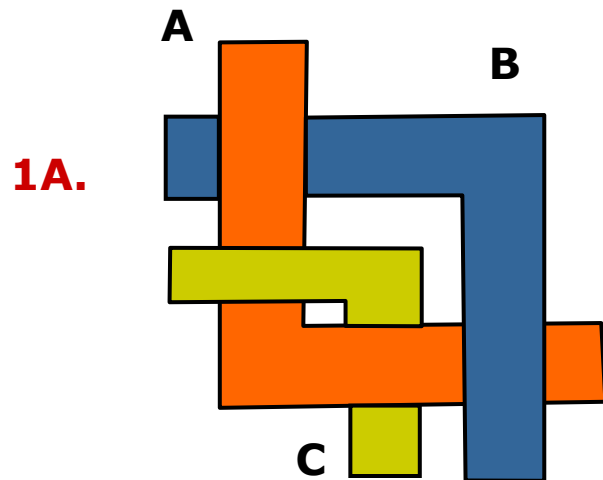
1B. 2 cilindri, 1 cub si 2 sfere.

2A. 2 sfere si 3 cilindri.

2B. 3 cuburi si 2 cilindri.

Subiect 2

Explicati decuparea in planul de proiectie a urmatoarelor poligoane prin algoritmul Weiler:

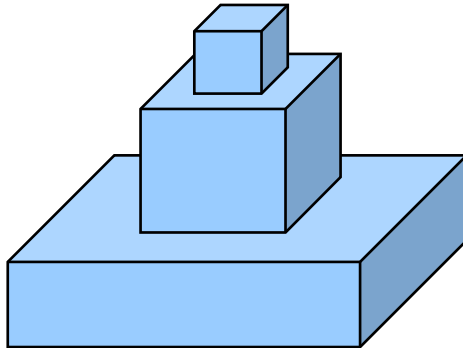


Subiect 3

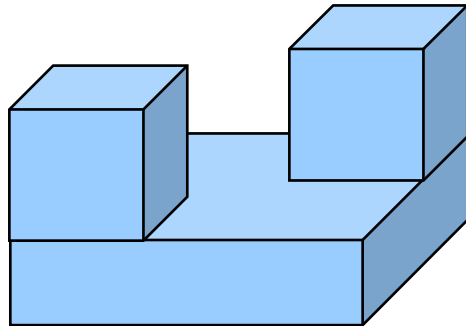
Exemplificati modelul Octree pentru obiectele date in urmatoarea figura.

Observatie: Se impune forma si pozitia relativa a obiectelor. Este la libera alegere dimensiunea lor.

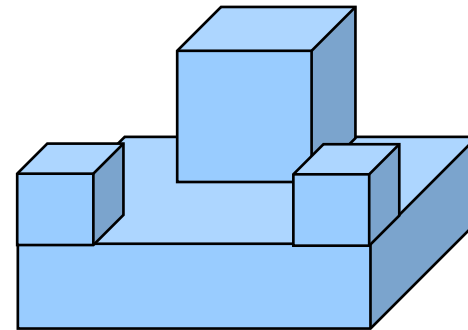
1A.



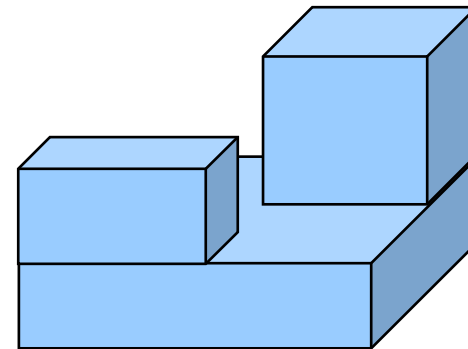
1B.



2A.



2B.

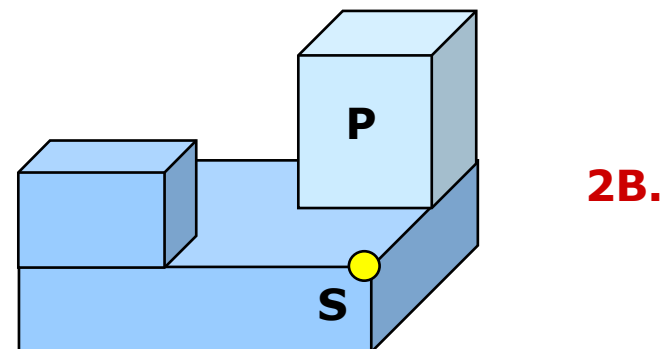
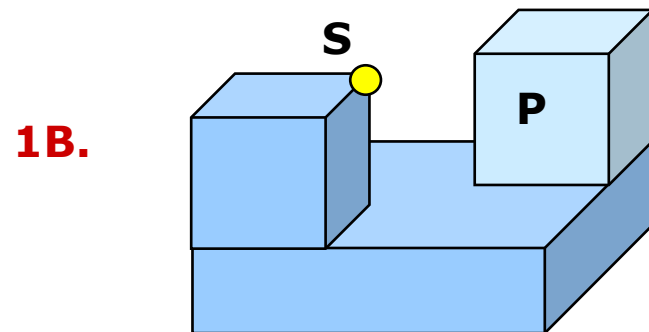
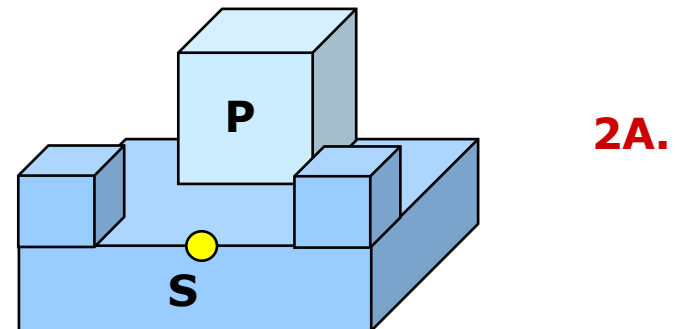
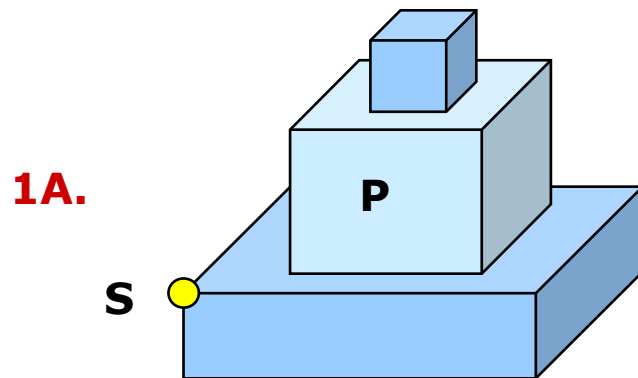


Subiect 4

Explicati secventa de transformare a unui obiect 3D din spatiul de vizualizare in spatiul ecran. Care sunt avantajele si dezavantajele acestei transformari.

Subiect 5

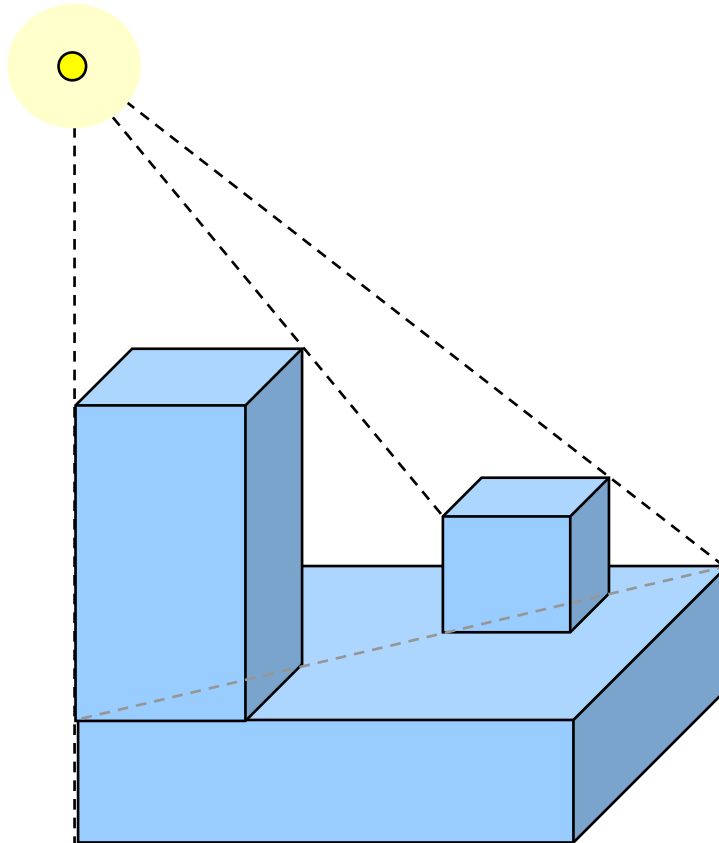
Explicati calcularea intensitatii luminoase folosind modelul de reflexie locala Phong, pentru urmatoarele obiecte. Detaliati calcularea incrementala a intensitatii luminoase prin metodele Gouraud si Phong pentru poligonul P, daca sursa de lumina punctiforma este in pozitia S data in figura:



Subject 6

Exemplificati si explicati calcularea umbrelor pentru obiectele din figura prin urmatorul algoritm :

- 1A.** Z-Buffer de umbra
(Shadow Z-Buffer)
- 1B.** Volume de umbra
(Shadow volumes)
- 2A.** Transformari
- 2B.** Scanline cu umbre
(Shadow scanline)



Subiect 7

Exemplificati si explicati tehnica inversa de mapare a unei texturi prin doua faze, pentru urmatorul caz, daca obiectul este un cilindru:

1A. S: T' este o sfera, O: normala la suprafata

1B. S: T' este un cub, O: normala la obiect

2A. S: T' este un cub, O: raza vizuala

2B. S: T' este un cilindru, O: raza vizuala

Subiect 8

Exemplificati si explicati algoritmul ray tracing pentru obiectele date la Subiectul 1 si doua surse de lumina. Primele 3 obiecte din lista sunt transparente, iar ultimele 2 sunt opace.

Subiect 9

Explicati si comparati transformarile dintr-un pipeline grafic clasic cu cele ale unui GPU. Definiti si explicati Arhitcetura Shader Unificat (Unified Shader Architecture).

Subiect 10

Explicati si comparati secventele grafice pipeline "sort first" si "sort last" pentru obiectele date la Subiectul 2, daca in final se obtine aceeasi imagine ca in figura.